Quiz 8

Friday, October 30 — linear dynamical systems (Week 9)

Last updated: October 12, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers the week before only: Dynamics, Simulation, and Through time (Week 9). Same-day Cost and Lab 3 are **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) · Lachniet Octave PDF · also on Resources.

Schedule

Core

Fluency (complete prompts — practice these by hand)

C8.1. A campus population has state $x_t = \begin{bmatrix} c_t \\ o_t \end{bmatrix}$ (on campus, off campus) with

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 1000 \\ 0 \end{bmatrix}.$$

Compute $x_1 = Ax_0$ by hand. In one sentence, what do the two entries of x_1 mean?

C8.2. With the same A and x_1 from C8.1, compute $x_2 = Ax_1$.

C8.3. True or false, with a one-line reason: for $x_{t+1} = Ax_t$, we always have $x_3 = A^3 x_0$.

C8.4. Let $A = \mathrm{diag}(1.2,\, 0.5)$ and $x_0 = (5, 4)$.

(a) Compute $x_1 = Ax_0$.

(b) After many steps, which component dominates, and does it grow or decay?

C8.5. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $x_0 = (1, 0)$. Compute x_1 and x_2. Describe the long-run behavior in one sentence (no eigenvalues required).

C8.6. Two towns exchange residents each year: - 90% of town A stays in A; 10% moves to B. - 80% of town B stays in B; 20% moves to A.

Write the transition matrix A so that $x_{t+1} = Ax_t$ with $x = \begin{bmatrix} a \\ b \end{bmatrix}$ (population in A, then B). Compute x_1 if $x_0 = (100, 50)$.

C8.7. Someone writes a loop `for t = 1 to 10: x = A*x` but never stores the initial x_0. How many states after x_0 does the loop produce? What should they add if they also want x_0 recorded?

C8.8. True or false, with a one-line reason: if you change x_0 but keep the same A, the long-run pattern of x_t must stay exactly the same for every t.

C8.9 (software sense). Octave/Python reports, after iterating $x \leftarrow Ax$ twenty times from $x_0 = (1000, 0)$ with the campus matrix in C8.1, that $x_{20} \approx (500, 500)$. What one-sentence observation would you write down for later (Week 13), without naming eigenvalues yet?

C8.10 (cumulative, light). Dimensions: if A is 2×2 and $x_t \in \mathbb{R}^2$, what are the dimensions of Ax_t? Of A^5?

Textbook Core

- B 9.1, 9.2, 9.3 (linear dynamical systems / simulation ideas used in class)
- L Ch. 2: matrix–vector products and simple loops as used in Lab 4 prep

Stretch

Simulate five steps of a different 2×2 A by hand; compare two initial states after many iterations and state what is surprising.

What a quiz item might change

The transition matrix entries, the initial state, asking for x_2 instead of x_1, or a short loop / software-output interpretation.